Data Structures

Recursion (Advanced)

Slides from Prantik Paul [PKP]

Recursive Definition (Binary Search)

```
if the array is empty (if l > r that is):
  return false
else:
  Find the position of the middle element: mid = (l + r)/2
  If key == data[mid], then return true
  If key > data[mid], the search the right half data[mid+1..r]
  If key < data[mid], the search the left half data[l..mid-1]
```

Recursive Definition (Binary Search)

```
                          -
                          | false                       if l > r
                          | true                        if k = a[mid]
contains(a,l,r,k)  =      |
                          | contains(a,mid+1,r,k)  if k > a[mid]
                          | contains(a,l,mid-1,k)  if k < a[mid]
                          -
```

Recursive Programming (Binary Search)

```
def contains(arr, left, right, key):
  if left > right:
    return False    #base case
  else:
    mid=(left+right)//2
    if key==arr[mid]:
      return True   #base case
    elif key > arr[mid]:
      return contains(arr, mid+1, right, key) #recursive part
    else:
      return contains(arr, left, mid-1, key)  #recursive part
```

Recursive Programming (Find Maximum - Binary)

```
def maximum(a,b):
  return a if a>=b else b

def findMax(arr, left, right):
  if left == right:
    return arr[left]    #base case
  else:
    mid = (left+right)//2
    maxLeftHalf=findMax(arr, left, mid)     #recursive part
    maxRightHalf=findMax(arr, mid+1, right) #recursive part
    return maximum(maxLeftHalf, maxRightHalf)
```

Recursive Definition (Exponentiation Efficient)

$$
a^n = \begin{cases} 1 & n = 0 \\ a^{n/2} \times a^{n/2} & n \text{ is even} \\ a^{(n-1)/2} \times a^{(n-1)/2} \times a & n \text{ is odd} \end{cases}
$$

Recursive Programming (Exponentiation Efficient)

```
def exp(a, n):
  if n == 0:
    return 1    #base case
  elif n % 2 == 0:
    return exp(a, n/2) * exp(a, n/2)                #recursive part
  else:
    return exp(a, (n-1)/2) * exp(a, (n-1)/2) * a    #recursive part
```

Recursive Programming (Exponentiation Efficient)

```
def exp(a, n):
  if n == 0:
    return 1     #base case
  elif n % 2 == 0:
    temp = exp(a, n/2)          #recursive part
    return  temp * temp
  else:
    temp = exp(a, (n-1)/2)      #recursive part
    return  temp * temp * a
```

Recursive Programming (Problems)

- **Inefficient Recursion**

1. **Inefficient recursion:**

The recursive solution for Fibonacci numbers outlined in these notes shows massive redundancy, leading to very inefficient computation. The 1st recursive solution for exponentiation also shows how redundancy shows up in recursive programs. There are ways to avoid computing the same value more than once by caching the intermediate results, either using Memoization (a top-down technique — see next topic), or Dynamic Programming (a bottom-up technique — survive this semester to enjoy it in the next one).

Recursive Programming (Problems)

- **Space for Activation Frames**
- **Infinite Recursion**

2. **Space for activation frames:**
Each recursive method call requires that it's activation record be put on the system call stack. As the depth of recursion gets larger and larger, it puts pressure on the system stack, and the stack may potentially run out of space.

3. **Infinite recursion:**
Ever forgot a base case? Or miss one of the base cases? You end up with infinite recursion, since there is nothing stopping your recursion! Whenever you see a Stack Overflow error, check your recursion!

Recursive Programming (Advanced)

- Top-Down Approach
- Redundancies
- Memoization

Recursive Programming (Advanced)

To develop a recursive algorithm or solution, we first have to define the problem (and the recursive definition, often the hard part), and then implement it (often the easy part). This is called the **top-down** solution, since the recursion opens up from the top until it reaches the base case(s) at the bottom.

Once we have a recursive algorithm, the next step is to see if there are **redundancies** in the computation— that is, if the same values are being computed multiple times. If so, we can benefit from **memoizing** the recursion. And in that case, we can create a memoized version and see what savings we get in terms of the running time.

Recursive Programming (Steps)

Recursion has certain overhead costs that may be minimized by transforming the memoized recursion into an iterative solution. And, finally, we see if there are further improvements that we can make to improve the time and space complexity.

Recursive Programming (Steps)

1. Write down the recursion,
2. Implement the recursive solution,
3. Memoize it,
4. Transform into an iterative solution, and finally
5. Make further improvements.

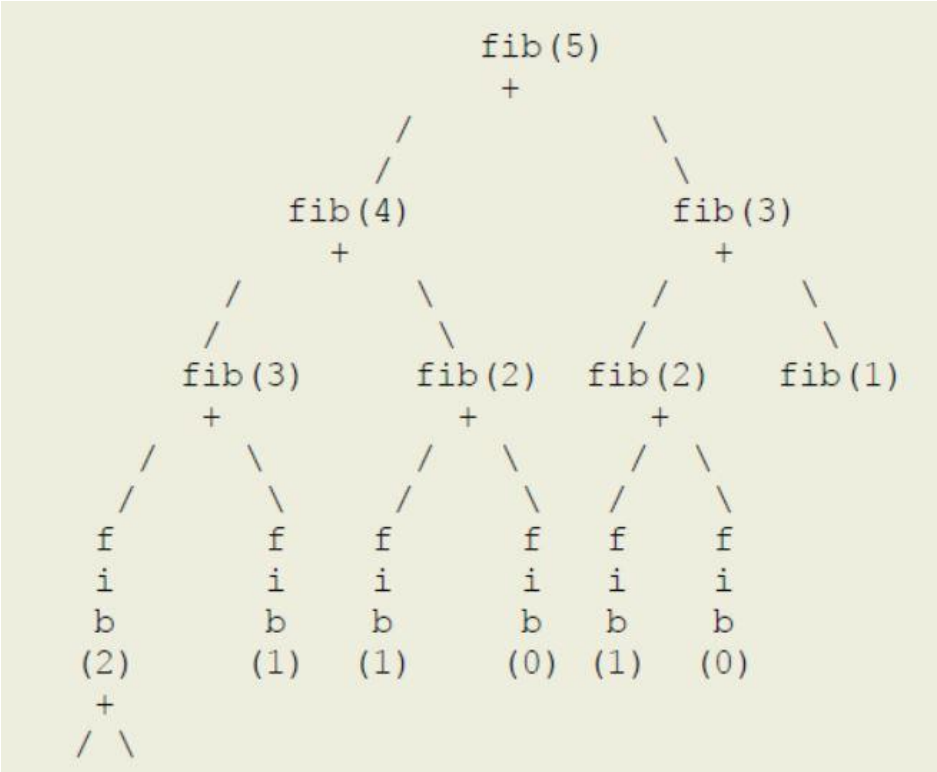
Recursive Programming (Fibonacci : Step 1)

$$
fib(n) = \begin{cases} n & \text{if } n < 2 \\ fib(n-1) + fib(n-2) & n \geq 2 \end{cases}
$$

Recursive Programming (Fibonacci : Step 2)

```
def fib(n):
  if n < 2:
    return n    #base case
  else
    return fib(n-1) + fib(n-2)    #recursive part
```

Recursive Programming (Fibonacci : Step 3)



Recursive Programming (Fibonacci : Step 3)

memoization trades space for time

Linear Array. Size : n+1

Recursive Programming (Fibonacci : Step 3)

```
def M_fib(n):
  # Assume that we have some "global" array (also called a table)
  # F with n+1 capacity. Why can F not be a local variable?

  if n < 2:
    return n    #base case
  else:

    #Compute (and save) if it's not already computed

    if (F[n] is empty)  # <<<< NOTE PSEUDOCODE!
      F[n] = M_fib(n-1) + M_fib(n-2)    #recursive part

    # Now just return the computed (and saved) value.
    return F[n]
```

Recursive Programming (Fibonacci : Step 3)

```
def fib(n):
  # Create and initialize the table.
  F = [-1]*(n+1)

  #Now we can call M-Fib with this extra parameter "F".
  return M_fib(n, F)

def M_fib(n, F):
  #The table "F" is being passed as a parameter

  if n < 2:
    return n    #base case

  else:
    #Compute (and save) if it's not already computed.
    if F[n] == -1:
      F[n] = M_fib(n-1, F) + M_fib(n-2, F)    #recursive part

    #Now just return the computed (and saved) value.
    return F[n]
```

Recursive Programming (Fibonacci : Step 4)

```
def fib(n):
  F=[None]*(n+1)  #The table to store computed values
  F[0]=0          #The base case for n = 0
  F[1]=1          #The base case for n = 1

  for i in range(2,n+1):
    F[i]=F[i-1]+F[i-2]
  return F[n]
```

Recursive Programming (Fibonacci : Step 5)

```
def fib(n):
  f_2 = 0     #The (n-2)th value
  f_1 = 1     #The (n-1)th value
  f = n
  #The result f is initialized to n (why? So that it works when n is 0 or 1).

  for i in range(2, n+1):
    f = f_1 + f_2

    #Now update the f_1 and f_2 for the next iteration (if any).
    f_2 = f_1
    f_1 = f

  return f
```